SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE CODE: ITA 0614 COURSE NAME: MACHINE LEARNING

COURSE OUTCOME COVERED

CO2: *Neural Network Representation, Perceptron Model, Multilayer Perceptrons, Backpropagation Algorithm, Deep Neural Networks, Genetic Algorithms, Hypothesis Space Search, Genetic Programming, Models of Evaluation and Learning (BL1, BL2)*

Assessment Tool Weightage: 40%

QUESTIONS: 15 **Total Marks:25**

Annexure A

Analytical Problem Solving

Code of Conduct:

I Variganji Sumanth Babu / 192524447, certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



Annexure A

Short Answer Test

1, A simple perceptron model is used to classify whether a student passes or fails based on two input features, namely study hours and attendance. The weights associated with these inputs are 0.6 and 0.4 respectively, and the bias value is –0.5. For a given student with study hours equal to 2 and attendance equal to 1, compute the net input to the perceptron and apply the step activation function to determine the final output. Based on the result, conclude whether the student passes or fails. (5 Marks)

2. In a neural network, a single neuron is trained using the backpropagation learning rule. The initial weight is 0.5 and the learning rate is 0.1. For an input value of 2, the actual output produced by the neuron is 0.6, whereas the target output is 1. Calculate the error in the output and determine the weight update using the delta rule. Hence, compute the new updated weight after one iteration of training. (5 Marks)

3. A deep learning model is developed for disease prediction and its performance is evaluated using a confusion matrix. The model yields 180 true positives, 150 true negatives, 20 false positives, and 50 false negatives. Using these values, compute the accuracy, precision, sensitivity (recall), and specificity of the model. Based on the calculated results, discuss whether the model is reliable for medical diagnosis, considering the importance of minimizing diagnostic errors. (5 Marks)

4.A Genetic Algorithm is applied to maximize the function $f(x) = x^2$ using an initial population consisting of the values 2, 4, 6, and 8. Determine the fitness value for each individual in the population and select the best candidates based on their fitness scores. Perform a crossover operation by exchanging the least significant bits of the selected individuals to generate new offspring. Additionally, explain the role of mutation in improving the diversity and effectiveness of the solution in Genetic Algorithms. (5 Marks)

5. During the training of a deep neural network model, the training and validation accuracies are observed over multiple epochs. The training accuracy increases steadily from 60% to 95%, while the validation accuracy initially increases from 58% to 72% but then decreases to 68% in later epochs. Analyse this trend and identify the issue affecting the model after a certain point in training. Explain why this issue occurs in neural networks and suggest two effective techniques to address it. Also, discuss the significance of validation accuracy in evaluating the performance of a learning model. (5 Marks)

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	**25**	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	**25**	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	**20**	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	**15**	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	**15**	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand

27-10-2024 COD - Assement Tool 1 - Analytical Problem Solving

Vaziganji Sumanth Babu
192524447

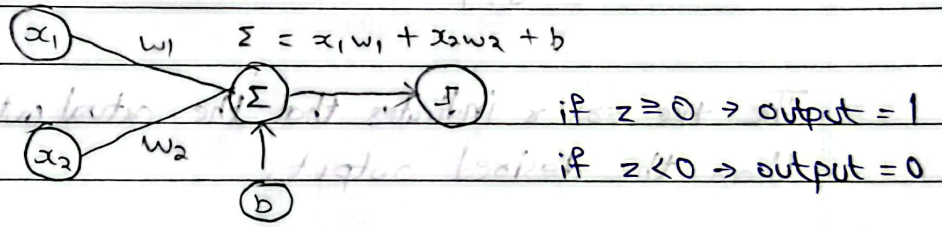
Q1. A simple perceptron classifies whether a student passes or fails based on study hours and attendance. The weights are 0.6 and 0.4, and the bias is -0.5. For a student with study hours = 2 and attendance = 1. Calculate the net input and apply the step activation function.

A perceptron calculates its net input using,

$$Net = w_1x_1 + w_2x_2 + b$$

$x_1 = 2$; $x_2 = 1$; $w_1 = 0.6$; $w_2 = 0.4$; $b = -0.5$

$$Net = (0.6 \times 2) + (0.4 \times 1) - 0.5$$
$$= 1.2 + 0.4 - 0.5$$
$$Net = 1.1 //$$



Apply step activation function,

$$f(Net) = \begin{cases} 1; & Net \geq 0 \\ 0; & Net < 0 \end{cases}$$

$1.1 \geq 0$

Output = 1

∴ Student PASSES

The perceptron structure consists of weighted inputs, a bias, a weighted summation and an activation function.

Q2. A single neuron is trained using the backpropagation learning rule. Initial weight = 0.5, learning rate = 0.1, input = 2, actual output = 0.6, and target output = 1. Calculate the error, weight update and ~~tret~~ net weight.

Given,

$w = 0.5$

$\eta = 0.1$

$x = 2$

Output = 0.6

Target = 1

Error = Target − Actual

$= 1 - 0.6$

$= 0.4$

The +ve error indicates that the actual output is lower than the desired output.

$\Delta w = \eta \times \text{Error} \times x$

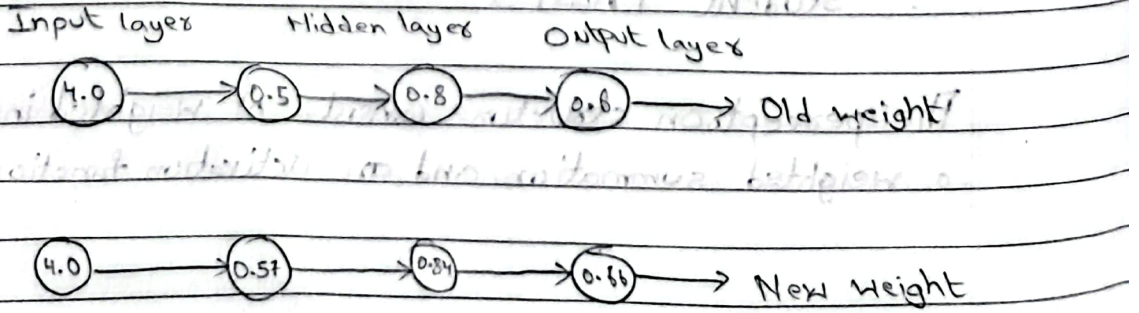
$= 0.1 \times 0.4 \times 2$

$\boxed{\Delta w = 0.08}$

$w_{new} = w_{old} + \Delta w$

$= 0.5 + 0.08$

$\boxed{w_{new} = 0.58}$



classmate
Date
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3. A deep learning model is developed for disease prediction and its performance is evaluated using a confusion matrix. The model yields 180 true positives, 150 true negatives, 20 false positives and 50 false negatives. Using these values, compute the accuracy, precision, sensitivity (recall), and specificity of the model. Based on the calculation results, discuss whether the model is reliable for medical diagonosis, considering the importance of minimizing diagnostic errors.

		Negative	Positive	
Negative		TN	FP	Specificity $\frac{TN}{(TN+FP)}$
Positive		FN	TP	Recall $\frac{TP}{(TP+FN)}$
		Negative Predictive value $\frac{TN}{(TN+FN)}$	Precision $\frac{TP}{(TP+FP)}$	Accuracy $\frac{(TP+TN)}{(TP+FP+TN+FN)} \times 100$

Given,

Actual/Predicted	+ve	-ve
+ve	TP = 180	FN = 50
-ve	FP = 20	TN = 150

1) Accuracy $= \frac{TP + TN}{TP + TN + FP + FN} \times 100$

$= \frac{180 + 150}{400} \times 100$

$= \frac{330}{400} \times 100 = 82.5\ \%$

Accuracy = 82.5 %

2) Precision = $\frac{TP}{TP + FP} \times 100$

$= \frac{180}{180+20} \times 100$

$= \frac{180}{200} \times 100$

Precision = 90 %

∴ This means that 90 % of the patients predicted as having the disease actually have it.

3) Sensitivity / Recall :

Sensitivity = $\frac{TP}{TP + FN} \times 100$

$= \frac{180}{180+50} \times 100 \Rightarrow \frac{180}{230} \times 100$

Sensitivity = 78.26 %

4) Specificity = $\frac{TN}{TN + FP} \times 100$

$= \frac{150}{150+20} \times 100 \Rightarrow \frac{150}{170} \times 100$

Specificity ≈ 88.24 %

Conclusion: The model performs fairly well, but its sensitivity should be improved before it is relied upon for medical diagonis.

Q4. Genetic Algorithm

A genetic algorithm is used to maximize: $f(x) = x^2$
Initial population: {2, 4, 6, 8}
Find the fitness value of each individual, select the best candidates, perform crossover by exchanging least significant bits, and explain mutation.

A Genetic Algorithm is an optimization technique inspired by natural evolution. It generally follows the process.

Step 1: Calculate the fitness

The fitness function is: $f(x) = x^2$

		Binary
For $x = 2$:	$f(2) = 2^2 = 4$	0010
For $x = 4$:	$f(4) = 4^2 = 16$	0100
For $x = 6$:	$f(6) = 6^2 = 36$	0110
For $x = 8$: (Best individual)	$f(8) = 8^2 = 64$ (Fitness)	1000

Step 2: Selection

The two strongest candidates select 8 and 6.

Step 3: Crossover

Using a two-bit crossover from the least significant side:

Parent 1 = 10 | 00 (8)
Parent 2 = 01 | 10 (6)
↑ exchange last 2 bits

After crossover:

Offspring 1 = 10 | 10 = 1010 = 10
Offspring 2 = 01 | 00 = 0100 = 4

Fitness values: $f(10) = 10^2 = 100$; $f(4) = 4^2 = 16$

Therefore, the new best candidate can be:

$x = 10$ with $f(10) = 100$

Step 4: Mutation

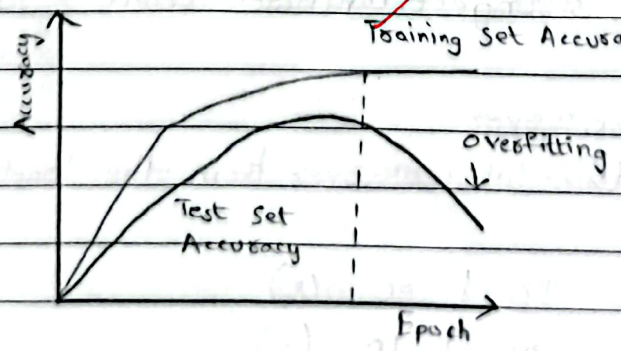
Before mutation:

1010 $\xrightarrow{\text{flip one bit}}$ 1000

∴, The next individual 10 has: $f(10) = 100$

Q5. Training vs Validation Accuracy / Overfitting.

During training of a deep neural network, training accuracy increases from 60 % to 95 %. Validation accuracy initially increases from 58 % to 72 %, but later decreases to 68 %. Analyze the trend, suggest two techniques to address it, and discuss the importance of validation accuracy.



Training Stage	Training Accuracy	Validation Accuracy
Initial	60 %	58 %
Improving	↑	↑
Best validation point	95 %	72 %
Later epochs	95 %	68 %

Technique 1: Early stopping

Early stopping stops training when validation performance stops improving.

Here, the best validation accuracy is: 72 %

Technique 2: Dropout

Dropout randomly deactivates a fraction of neurons during training.

Importance of Validation Accuracy:

1. Detecting overfitting
2. Selecting the appropriate number of training epochs.
3. Comparing different models.
4. Selecting hyperparameters.
5. Estimating how well the model may generalize to new data.

∴, The model is suffering from Overfitting.